

Math 4610: Fundamentals of Numerical Analysis

Practice Exam

Note: Show all work to receive full credit.

1. Find $\max_{a < x < b} |f(x)|$ for the function $f(x) = e^x + 3x$, on the interval $x \in [0, 1]$.
2. (a) Find the second Taylor polynomial $P_2(x)$ for the function $f(x) = xe^x + x$, about $x_0 = 0$.
(b) Find a bound for the error $|P_2(x) - f(x)|$ in using $P_2(x)$ to approximate $f(x)$ on the interval $[0, 1]$.
3. Use the 64-bit long real format to find the decimal equivalent of the machine number:

0 00000000011 1100000000000000...00000

4. Use the three-digit rounding arithmetic to perform the calculation for

$$(-10\pi + 3).$$

5. Suppose we are asked to solve the quadratic equation

$$1.002x^2 - 1000.0x + 0.25 = 0$$

by four-digit rounding arithmetic. Let $a = 1.002$, $b = -1000.0$, and $c = 0.25$. We are given the following four formulas as

(a) x_1 by

$$\frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad \text{or by} \quad \frac{-2c}{b + \sqrt{b^2 - 4ac}}.$$

(b) x_2 by

$$\frac{-b - \sqrt{b^2 - 4ac}}{2a}, \quad \text{or by} \quad \frac{-2c}{b - \sqrt{b^2 - 4ac}}.$$

Find the most accurate formulas to compute x_1 and x_2 and explain why (Note: you do not need to actually compute x_1 and x_2).

6. Use the bisection method to find P_2 for $f(x) = \sqrt{x} - \cos x$ on the interval $x \in [0, 1]$ (Note: P_2 is the approximate solution at the second iteration).
7. If we use the fixed-point iteration to find an approximation to the fixed point that is accurate to within 10^{-2} for $g(x) = \pi + 0.5 \sin(x)$ on the interval $x \in [0, \pi]$, how many iterations do we need?

8. Let $f(x) = -x^3 - \cos x$. Use Newton's method to find P_2 with $P_0 = -1$ (Note: P_2 is the approximate solution at the second iteration).
9. (a) Use the appropriate Lagrange interpolating polynomial of degree two $P_2(x)$ to approximate $f(0.22)$ if $f(0.1) = 0.665$, $f(0.2) = 0.8$, $f(0.3) = 1.8$, and $f(0.4) = 0.25$.
 (b) Suppose we know that $|f^{(3)}(x)| < 5$ on the interval containing 0.1, 0.2, 0.3, and 0.4. What is the error $|f(0.22) - P_2(0.22)|$ made by the interpolation?
 (Note: You do not need to find the actual value of $f(0.22)$; instead, you only need to write down the interpolating polynomial.)
10. (a) Use proper two-point difference formula to determine the missing entry in the following table.

x	$f(x)$	$f'(x)$
0.5	1.9	
0.6		2.2

- (b) Use the most accurate three-point formula to determine $f'(0.6)$ based on the following table.
 Use the central difference formula to determine $f''(0.6)$.

x	$f(x)$
0.5	1.9
0.6	2.3
0.7	1.5

11. Find the degree of accuracy of the quadrature formula

$$\int_0^a f(x) dx = \frac{a}{2}f(0) + \frac{a}{2}f(a).$$

Here $a > 0$.

12. Approximate the following integral using Simpson's rule.

$$\int_{-0.5}^{1.5} x \ln(x+2) dx.$$

13. Find an error bound for

$$\int_{-0.5}^{1.5} x \ln(x+2) dx$$

approximated by Simpson's rule.